

LANCE KAVALSKY

Postdoctoral Researcher, University of Wisconsin-Madison

📍 Madison, WI | ✉ kavalsky@wisc.edu | 🔗 LinkedIn | 🌐 lancekavalsky.github.io

EDUCATION & TRAINING

Postdoctoral Researcher in Chemical Engineering University of Wisconsin-Madison Advisor: Prof. Manos Mavrikakis	2024-present
PhD in Mechanical Engineering University of Michigan (continued from Carnegie Mellon University) Advisor: Prof. Venkatasubramanian Viswanathan Thesis: Electrochemical Manufacturing: Advances in Design and Accelerated Materials Discovery	2019-2024
MS in Mechanical Engineering Carnegie Mellon University Advisor: Prof. Venkatasubramanian Viswanathan	2019-2023
MASc in Materials Science and Engineering University of Toronto Advisor: Prof. Chandra Veer Singh	2017-2019
HBSc in Physics with High Distinction University of Toronto	2013-2017

PUBLICATIONS

* indicates equal author contribution |  ORCID |  Google Scholar

17. **L. Kavalsky** and M. Mavrikakis. "Adsorption of CO on gold: effect of coverage and surface deformations" *Catal. Sci. Technol.* (2026) (in press)
16. K.K. Li*, **L. Kavalsky***, M. Figueras-Valls, Y. Ding, M. Mavrikakis, Y. Xia. "What Makes Au Nanospheres Superior to Octahedral and Cubic Counterparts for the Deposition of a Pt Monolayer Shell?" *J. Am. Chem. Soc.*, 147, 29, 25273-25286 (2025)
15. V. Azumah, **L. Kavalsky**, V. Viswanathan. "Lithium Nitride (Li₃N) Formation in Lithium-Mediated Electrochemical Ammonia Synthesis Can Be Enhanced with the Right Proton Donor" *J. Catal.*, 450, 116250 (2025)
14. V. Azumah, **L. Kavalsky**, V. Viswanathan. "Proton Donors Influence Nitrogen Adsorption in Lithium-Mediated Electrochemical Ammonia Synthesis" *J. Phys. Chem. C*, 129, 5, 2488-2501 (2025)
13. M.S. Johnson, D. Farina, **L. Kavalsky**, *et. al.*. "Automatic Generation of Chemical Mechanisms for Electrochemical Systems: Solid Electrolyte Interphase Formation in Lithium Batteries" *J. Phys. Chem. C*, 129, 28, 12667-12678 (2025)
12. **L. Kavalsky** and V. Viswanathan. "Electrowinning for room-temperature ironmaking: Mapping the electrochemical aqueous iron interface" *J. Phys. Chem. C*, 128, 35, 14611-14620 (2024) (part of Jens K. Nørskov Festschrift virtual special issue)
11. **L. Kavalsky**, V.I. Hegde, B. Meredig, V. Viswanathan. "A Multiobjective Closed-loop Approach Towards Autonomous Discovery of Electrocatalysts for Nitrogen Reduction" *Digital Discovery*, 3, 999-1010 (2024)
10. **L. Kavalsky***, V.I. Hegde*, E. Muckley, M.S. Johnson, B. Meredig, V. Viswanathan. "By how much can closed loop frameworks accelerate computational materials discovery?" *Digital Discovery*, 2, 1112-1125 (2023)
9. R. Patil, M. Kaur, S. House, **L. Kavalsky**, *et. al.*. "Reversible Alkaline Hydrogen Evolution and Oxidation Reactions Using Ni-Mo Catalysts Supported on Carbon" *Energy Adv.*, 2, 1500-1511 (2023)
8. E. Annevelink*, R.C. Kurchin*, E. Muckley, **L. Kavalsky**, *et. al.*. "AutoMat: Accelerated Computational Electrochemical systems Discovery" *MRS Bulletin*, 47, 1036-1044 (2022)

7. L.M. Tsiverioti, **L. Kavalsky**, V. Viswanathan. "Robust Analysis of $4e^-$ vs $6e^-$ Reduction of Nitrogen on Metal Surfaces and Single-Atom Alloys" *J. Phys. Chem. C*, **126**, 31, 12994-13003 (2022)
6. **L. Kavalsky** and V. Viswanathan. "Robust Active Site Design of Single-Atom Catalysts for Electrochemical Ammonia Synthesis" *J. Phys. Chem. C*, **124**, 42, 23164-23176 (2020)
5. S. Mukherjee, **L. Kavalsky**, K. Chattopadhyay, C.V. Singh. "Dramatic improvement in the performance of graphene as Li/Na battery anodes with suitable electrolytic solvents" *Carbon*, **161**, 570-576 (2020)
4. **L. Kavalsky**, S. Mukherjee, C.V. Singh. "Compression induced resistance of singlet oxygen dissociation on phosphorene" *Phys. Rev. Materials*, **4**, 021001 (2020) (Editor's Suggestion highlight)
3. **L. Kavalsky**, S. Mukherjee, C.V. Singh. "Phosphorene as a Catalyst for Highly Efficient Nonaqueous Li-Air Batteries" *ACS Appl. Mater. Interfaces*, **11**, 1, 499-510 (2019)
2. S. Mukherjee, **L. Kavalsky**, K. Chattopadhyay, C.V. Singh. "Adsorption and diffusion of lithium polysulfides over blue phosphorene for Li-S batteries" *Nanoscale*, **10**, 45, 21335-21352 (2018)
1. S. Mukherjee*, **L. Kavalsky***, C.V. Singh. "Ultrahigh storage and fast diffusion of Na and K in blue phosphorene anodes" *ACS Appl. Mater. Interfaces*, **10**, 10, 8630-8639 (2018)

HONORS & AWARDS

NAM28 Kokes Award , North American Catalysis Society	2023
NAM27 Kokes Award , North American Catalysis Society	2022
NSERC CGS-D , Natural Sciences and Engineering Research Council of Canada	2020
Haultain Fellowship , University of Toronto	2018
Graduate Student Endowment Fund Award , University of Toronto	2017
New College In-Course Scholarship , University of Toronto	2014
New College Open Admission Scholarship , University of Toronto	2013

EXPERIENCE

Postdoctoral Researcher, University of Wisconsin-Madison 2024-Present

- Established physics-informed tests targeting specific chemical interactions for machine learning interatomic potential (MLIP) validation highlighting model to enable multiscale modeling of heterogeneous catalysts
- Unveiled a rounding effect in Au nanospheres by training and applying an MLIP with large-scale molecular dynamics simulations (~ 60 -100k atoms), guided by characterization experiments performed by collaborators
- For ORR catalysis, revealed the relationships between ligand and strain effects to explain the anion poisoning resistance of intermetallics observed by experimental collaborators through first-principles calculations
- Uncovered the relationship between surface deformation of gold and CO saturation coverage and benchmarked against infrared spectroscopy from literature

Graduate Research Assistant, University of Michigan 2023-2024

- Identified hydrogen evolution on step defects as a core obstacle to electrochemical ironmaking through computational modeling
- Engineered and open-sourced methodologies for closed-loop electrocatalyst discovery that pairs machine learning generated hypotheses with automated first-principles workflows
- Led the multidisciplinary DFT subgroup to discuss research updates and troubleshoot code/results

Graduate Research Assistant, Carnegie Mellon University 2019-2023

- Open-sourced a software for active-learning guided computational discovery of electrocatalysts, [AutoCat](#) and developed scripts for high-throughput first-principles calculations
- Demonstrated that computational closed-loop frameworks could accelerate materials discovery by 10 - $20\times$ through rigorous acceleration benchmarking
- Autonomously explored the pareto front for electrocatalysts (defined by proxies for activity, cost, and stability) by proposing and implementing a multiobjective scoring scheme for closed-loop discovery

Graduate Research Assistant, University of Toronto 2017-2019

- For Na-ion and K-ion batteries, assessed storage and diffusion over 2D phosphorus through first-principles analysis
- Identified strain engineering as an approach to suppress oxygen dissociation on 2D phosphorus toward improved stability via metadynamics-accelerated ab-initio MD calculations

PATENTS AND PATENT APPLICATIONS

1. US Patent Application PCT/US2024/050004 “Electrochemical Synthesis of Direct Reduced Metal and Metal-derived Compounds” Inventors: **Kavalsky, L.** (Ann Arbor, MI), Viswanathan, V. (Ann Arbor, MI)

PRESENTATIONS

Oral Talks

5. **L. Kavalsky**, M. Mavrikakis. “Modeling Au Surfaces in Reactive Environments via Machine-Learned Interatomic Potentials” *2025 MRS Fall Meeting*, Boston, MA (December 2025)
4. **L. Kavalsky**, X. Zhao, K. Sasaki, M. Mavrikakis. “Understanding Anion Adsorption in Low Pt-Loading N-Doped Intermetallic Fuel Cell Cathodes” *248th ECS Meeting, Chicago, IL (October 2025)*
3. (Invited) **L. Kavalsky**, M. Mavrikakis. “Developing Machine-Learned Interatomic Potentials for Au: A Gold Mine for Surface Science Insights” Hougén/CATHEX Symposium, Madison, WI (June 2025)
2. **L. Kavalsky**, V.I. Hegde, V. Viswanathan. “Autonomous Multiobjective Computational Discovery of NRR Electrocatalysts” *The 28th North American Catalysis Society Meeting*, Providence, RI (June 2023)
1. **L. Kavalsky**, V.I. Hegde, E. Muckley, L.M. Tsiverioti, V. Viswanathan. “A Generalizable Closed-Loop Framework for Accelerated Discovery of Electrochemical Nitrogen Reduction Catalysts” *The 27th North American Catalysis Society Meeting*, New York City, NY (May 2022)

Poster Presentations

3. **L. Kavalsky**, X. Zhao, K. Sasaki, M. Mavrikakis. “Investigating Anion Poisoning on N-doped L1₀ Intermetallic Fuel Cell Cathodes through First-Principles Modeling” *2025 MRS Fall Meeting*, Boston, MA (December 2025) (**MRS Best Poster Award Nominee**)
2. **L. Kavalsky**, V.I. Hegde, V. Viswanathan. “Accelerated Computational Discovery of Electrocatalysts via an Autonomous Multiobjective Workflow” *Accelerate Conference, Toronto, ON (August 2023)*
1. **L. Kavalsky**, V. Viswanathan. “Unraveling the Iron-Water Interface for Electrochemical Steel Production” *The 28th North American Catalysis Society Meeting*, Providence, RI (June 2023)

TEACHING EXPERIENCE

TA: Numerical Methods in Engineering , Carnegie Mellon University	Spring 2023
• Assisted running hands-on in-class exercises • Led an introductory C++ coding workshop and ran homework review sessions	
TA: Energy Storage Materials & Systems , Carnegie Mellon University	Fall 2021
• Held office hours to address student questions and cover supplementary course content • Led an introductory Python coding workshop	
TA: Computational Materials Design , University of Toronto	Fall 2018
• Conducted lab sessions which provided senior undergraduate and graduate students hands-on experience with Density Functional Theory and Molecular Dynamics techniques • Delivered a guest lecture on Ab-Initio Molecular Dynamics and Metadynamics	
Counsellor: Da Vinci Engineering Enrichment Program , University of Toronto	July 2017
• Provided teaching support in the Foundations of Quantum Mechanics course • Led a session on superconductivity	

ACADEMIC SERVICE

Reviewing

ACS Applied Energy Materials, Journal of Open Source Software